SPF: Background, Problem Space, and Approach

RF emitter localization with two-element software-defined radio arrays

Project SPF

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Abstract

SPF localizes radio-frequency emitters using pairs of cheap two-antenna software-defined radios (SDRs) mounted on mobile rovers and on a mechanically positioned wall array. This document motivates the project, maps the problem space, and describes the approach: large-scale real-world data collection, a careful physical model of what a two-element array can and cannot measure, and learned models plus filtering that recover bearing (and, with multiple receivers, position) despite ambiguity and hardware imperfection. A dedicated section walks step by step through the core mathematics: how the phase difference between two antennas encodes the angle of arrival, where the ambiguities come from, and how the same forward model powers both the neural networks' inputs and the dataset-quality audit.

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1 Motivation

Finding *where a radio transmission is coming from* is a classical problem with modern urgency: locating a lost drone by its video downlink, finding an interference source that is degrading a shared band, tracking wildlife tags, or navigating toward a beacon when GPS is unavailable. Commercial direction-finding gear solves this with large, calibrated, many-element antenna arrays and correspondingly large price tags.

SPF starts from the opposite end of the cost spectrum and asks:

> *How much localization capability can be extracted from the cheapest possible sensor — a single pair of antennas attached to a \$100–\$200 software-defined radio — if we are willing to move the sensor, collect a lot of real data, and learn the sensor's imperfections instead of engineering them away?*

Three observations make this question worth asking.

1. Two antennas are the minimal interferometer. A two-element array measures exactly one number per snapshot — the phase difference between its channels — and that number is a (nonlinear, ambiguous, noisy) function of the emitter bearing (Section 4). Everything the platform will ever know about direction must be squeezed out of this one degree of freedom, plus motion, plus time. It is the sharpest possible test bed for "sensing with learned models": the physics is simple enough to write down exactly, yet the real-world measurement deviates from that physics in ways that are hard to calibrate by hand (mount rotations, effective antenna spacing, gain-dependent phase inversions, automatic gain control, multipath).

2. Motion converts ambiguity into information. A stationary two-element array cannot distinguish front from back, and for wide antenna spacing cannot even pin down a unique bearing (Section 4.6). But a *moving* receiver– or a pair of receivers — sweeps its ambiguity structure through space; a filter that accumulates measurements over a trajectory can collapse the ambiguity. This trades hardware complexity for algorithmic complexity, which is exactly the trade a learned system should win.

3. Real data beats simulated calibration. Early experiments made it clear that the dominant error sources are not thermal noise but *systematics*: antennas mounted a few millimetres from their nominal positions, arrays labeled with the wrong spacing, compass biases that drift between calibration sessions, receiver gain control interacting with the RF front end. These are properties of each physical rig on each particular day. SPF therefore treats large-scale, ground-truth-annotated data collection — and the auditing of that data (Section 5.4) — as a first-class part of the project, on equal footing with model architecture.

2 Problem space

2.1 Task statement

Given streams of IQ samples from the two RX channels of one or more two-antenna SDR receivers, each annotated with the receiver's pose (position, heading) and, at training time, the emitter's ground-truth position, estimate:

1. **Bearing** θ of the emitter relative to each receiver (the *single-radio* problem);
2. **Fused bearing / position** using two radios on one platform or multiple platforms over time (the *paired* and *filtering* problems).

2.2 Hardware platforms

Wall array (controlled environment). A GRBL-driven 2-axis gantry moves an emitter across a wall while fixed two-antenna receivers (PlutoSDR class, and later BladeRF 2 on a desk rig) record. Ground truth is sub-millimetre from the stepper positions. This platform isolates the RF chain: with essentially perfect geometry, any residual between predicted and measured phase is a property of radios, antennas, mounts, and environment — which is what makes it the reference platform for the data audit.

Rovers (field environment). Skid-steer rovers carry receivers (and one carries the emitter) around a field; poses come from GPS and compass, so ground-truth bearing error of a few degrees is intrinsic (3 m GPS error at 15 m range is ≈ 0.2 rad). Multiple rovers record simultaneously and their captures are merged into paired datasets (TX rover $\times$ RX rover).

Radios and signals. AD9361-family 2×RX SDRs (PlutoPlus, BladeRF 2) sampling up to 30 MHz. Emitters are continuous-wave-like beacons (ESP32 "tone blaster" on 2.4 GHz Wi-Fi channels; analog video transmitters at 5.7–5.9 GHz; sub-GHz LoRa-class modules at 868/915 MHz). Configured antenna spacings range from $d/\lambda \approx 0.21$ to 1.55 across the fleet — deliberately including spacings well past the unambiguous limit of $d/\lambda = 1/2$ (Section 4.6).

2.3 What makes it hard

Ambiguity by construction. Front–back mirror symmetry always; grating-lobe aliasing whenever $d/\lambda > 1/2$. The sensor *by design* does not determine a unique bearing per snapshot.

Systematic hardware error. Mount rotations and mislabeled spacings (both have occurred and were later fixed by "data surgery"), effective spacing differing from configured spacing (antenna coupling, connector geometry), per-channel LO phase offsets.

Gain control interactions. The AD9361 runs fast-attack AGC; gain changes of 17–39 dB occur within datasets, and gain-index changes can invert channel polarity at the LNA-bypass boundary — a π phase jump in the measurement of interest.

Intermittent, contested spectrum. Duty-cycled emitters (burst beacons, ∼1–3% airtime in the extreme), other transmitters in-band (Wi-Fi, other 5.8 GHz video links), and sessions with recorded interference incidents.

Label noise on moving platforms. GPS + compass ground truth is itself noisy and era-dependent (compass calibration events shift the bias).

Scale and drift of the corpus. Thousands of capture sessions collected over ∼1.5 years, across hardware revisions, firmware versions, frequency bands, and collection sites. Quality is strongly era-dependent; some months are systematically bad. Training and evaluating on this corpus requires an explicit quality model of the corpus itself.

2.4 Why not classical array processing alone?

MUSIC/ESPRIT-style methods assume a calibrated steering model and more elements than sources. With two elements, one source, uncalibrated mounts, and gain-dependent polarity, the classical estimate collapses to $\hat{\theta} = \arcsin(-\hat{\varphi}\,\lambda/2\pi d)$ applied to a noisy, aliased $\hat{\varphi}$ — brittle exactly where our data lives. The physics still matters (it defines the feature space and the audit's forward model), but the mapping from measured phase statistics to bearing posterior is learned, and ambiguity resolution is deferred to fusion and filtering.

Does any of this survive burst-mode digital signals? Yes — the geometry does not care whether the emitter transmits continuously, and the phase measurement is *per sample*, not per second. Four facts make short-lived packets (Wi-Fi, telemetry beacons) workable, and two things genuinely change:

- **A single packet is thousands of measurements.** The inter-channel product $x_1[n]x_0^*[n]$ (Section 4.5) yields one phase sample per IQ sample. A $100\,\mu$s packet at 30 MHz sampling is ∼3,000 complex samples — ample for a tight per-burst $\hat{\varphi}$.
- **Modulation cancels.** Both antennas receive the *same* transmitted waveform $s(t)$; in $x_1x_0^*$ the modulation contributes $|s|^2$ and drops out of the phase, leaving the geometric term — valid whenever the inter-antenna delay $\tau \lesssim 0.2\,\text{ns}$ is far below a symbol time ($B\tau \approx 4 \times 10^{-3}$ even for 20 MHz Wi-Fi). A modulated burst is as good as CW *within* the burst.
- **No coherence needed between bursts.** Both RX channels share one LO, so the phase *difference* is stable even if absolute phase drifts arbitrarily between packets; each burst contributes an independent bearing measurement to the filter.
- **Carrier knowledge per burst suffices.** Frequency-agile emitters change λ, hence the φ-scale, per hop — the estimator needs the hop frequency (from the capture configuration or a per-burst frequency estimate), not continuity.

What does change: (i) **segmentation becomes part of the measurement** — windows without signal contain uniform-phase junk and must be detected and gated, which is why duty-cycle and empty-window statistics are first-class quantities in the dataset audit; and (ii) **AGC transients concentrate at burst onsets** — fast-attack gain slews exactly when the packet starts, distorting amplitudes (and potentially polarity) for the first microseconds. Both effects hit fixed-form classical estimators harder than they hit an approach built on gated window distributions, which can simply down-weight or skip the corrupted windows.

3 Signals background: how we get complex numbers out of an antenna

The math section ahead manipulates "complex baseband samples" as if that were the obvious thing a radio produces. It is not obvious. This section motivates it from scratch: what the antenna really measures, why we describe it with complex numbers, what I/Q means, and precise working definitions of the terms (narrowband, broadband, window, coherent) used everywhere else.

3.1 What the antenna actually measures

An antenna is a wire; the only thing it can give the receiver is a *real* voltage that wiggles in time, $v(t)$. For a 2.412 GHz emitter that voltage oscillates 2.4 *billion* times per second. Almost none of those wiggles are interesting individually — the information (and, for us, the geometry) rides on how the wiggle's *strength* and *timing* drift over microseconds, millions of times slower than the wiggle itself.

Any such signal can be written as

$$v(t) \;=\; A(t)\,\cos\big(2\pi f t + \psi(t)\big),$$

where f is the **carrier frequency** (the fast wiggle), $A(t) \geq 0$ is the **amplitude** or **envelope** (how strong the wiggle currently is), and $\psi(t)$ is the **phase** (how the wiggle is currently shifted relative to a perfect clock at frequency f). A and ψ change slowly compared to the carrier; they are "the signal". The pair (A, ψ) is what we actually want, sampled in time.

3.2 Why complex numbers: one arrow instead of two numbers

A pair (strength, angle) is naturally a single *arrow* in the plane — a complex number:

$$x(t) \;=\; A(t)\,e^{j\psi(t)} \qquad \text{(length = amplitude, direction = phase).}$$

This is not just notation. The two operations radio processing does constantly become trivial arithmetic on arrows:

- **Delaying** a narrowband signal by τ rotates its arrow by a fixed angle: $x(t-\tau) \approx x(t)\,e^{-j2\pi f\tau}$ (the envelope barely moves, the carrier phase shifts). Delay — which is geometry — becomes multiplication.
- **Comparing** two channels becomes one multiplication: $x_1 x_0^*$ has length $A_1 A_0$ and angle $\psi_1 - \psi_0$. The entire two-antenna measurement of Section 4 is this one product.

With real cosines both operations are trigonometric-identity soup; with arrows they are one-liners. That is the whole reason radios use complex numbers.

3.3 I/Q: how the hardware extracts the arrow

The receiver cannot see A and ψ directly; it recovers them by **mixing**: multiplying $v(t)$ by its own **local oscillator (LO)** — a clean internal tone at (nearly) the carrier frequency — in two copies, 90° apart:

$$\text{I branch:}\;\; v(t)\cdot 2\cos(2\pi f t) = A\cos\psi \;+\; \underbrace{A\cos(4\pi f t + \psi)}_{\text{at } 2f} \;\xrightarrow{\text{low-pass}}\; I = A\cos\psi,$$

$$\text{Q branch:}\;\; v(t)\cdot\big(-2\sin(2\pi f t)\big) = A\sin\psi \;+\; (\text{term at } 2f) \;\xrightarrow{\text{low-pass}}\; Q = A\sin\psi.$$

(The product-to-sum identity creates a slow term and a double-frequency term; the low-pass filter keeps the slow one.) The pair is exactly the arrow's Cartesian coordinates:

$$x = I + jQ = Ae^{j\psi}.$$

I is the *in-phase* component (relative to the LO cosine), **Q** the *quadrature* (90°-shifted) component. The SDR digitizes both at the **sample rate** f_s (up to 30 MHz here), producing the complex

baseband stream $x[n]$ used everywhere in this document. "Baseband" means the carrier has been divided out: a pure unmodulated carrier at exactly the LO frequency becomes a *constant* arrow.

Two hardware facts matter enormously for SPF:

- **Both RX channels of the AD9361 share one LO.** The arbitrary LO phase appears identically in ψ_0 and ψ_1 and cancels in $x_1 x_0^*$. This is why the *inter-channel phase difference* is physically meaningful and stable, even across signal bursts, while the *absolute* phase of either channel alone is meaningless.
- **The LO is never exactly the emitter's carrier.** A small frequency offset makes the arrow rotate slowly ("spin"); it is common to both channels and cancels in the difference, but it is why single-channel phase drifts and must be detrended in some analyses.

3.4 Term definitions used throughout

Carrier / baseband. The carrier is the fast oscillation at f; baseband is the slow complex signal $x(t)$ left after the carrier is mixed out.

Bandwidth B. How fast the arrow moves: the width of the frequency band occupied by $A(t)e^{j\psi(t)}$. A pure tone has $B \to 0$; a 20 MHz Wi-Fi packet has $B = 20\,\mathrm{MHz}$.

Narrowband (for our purposes). Two conditions, both comfortably true in SPF: (i) $B \ll f$ (the signal is a thin slice around the carrier), and (ii) — the one that matters for arrays — $B\tau \ll 1$, where $\tau \le d/c$ is the inter-antenna delay: the envelope does not change in the $\sim$0.2 ns it takes the wave to cross the array, so both antennas see *the same* A and ψ and differ only by the geometric rotation $e^{-j2\pi f\tau}$. Even Wi-Fi ($B\tau \approx 4 \times 10^{-3}$) passes easily.

Broadband / wideband. $B\tau$ not small, or B/f large: different frequency components inside the signal have different wavelengths, hence *different* inter-antenna phase shifts — a single φ no longer summarizes the geometry and processing must split the band (e.g. per-subcarrier). SPF's beacons do not operate here, but strong wideband interferers do.

Window. A contiguous run of N samples treated as one measurement unit; SPF computes phase/beamformer statistics per window, many windows per snapshot.

Snapshot / session. A snapshot is one capture buffer with one ground-truth pose; a session (dataset) is the full recorded trajectory of snapshots.

Coherent vs. incoherent combining. Coherent: add the complex arrows, *then* take magnitude/angle — signals aligned in phase reinforce (N arrows of unit length aligned give length N; random phases give $\sim\sqrt{N}$), so SNR grows with window length. Incoherent: add magnitudes or powers, discarding phase — robust when phase is unstable, but no phase gain. The coherent sum $\sum x_1 x_0^*$ of Section 4.5 is the coherent combine of the *difference* arrows.

SNR. Signal-to-noise power ratio within the processed band; window statistics turn per-sample SNR into per-window estimate quality.

With this vocabulary, the math section can be read literally: every "complex baseband sample" is an I/Q arrow, every phase difference is an angle between two arrows sharing one clock, and every "narrowband" invocation is the statement $B\tau \ll 1$ justified above.

4 The mathematics: from phase difference to angle

This section is deliberately slow. Everything SPF does — the beamformer features fed to the networks, the symmetry constraints in training, the particle filter's measurement model, and the dataset audit's forward model — is a consequence of one small piece of geometry, so it pays to derive it once, carefully, with signs and conventions pinned down.

4.1 Setup, conventions, and notation

Every symbol used repeatedly in this document is defined here, in bold at first use, and collected in Table 1.

Two receive **antennas $\boldsymbol{A_0}$ and $\boldsymbol{A_1}$** are separated by a **baseline of length $\boldsymbol{d}$** (metres, the "antenna spacing"). Place the midpoint of the baseline at the origin, with the baseline along the x-axis:

$$A_0 = \left(+\tfrac{d}{2},\, 0\right), \qquad A_1 = \left(-\tfrac{d}{2},\, 0\right).$$

The *boresight* (broadside) direction is the $+y$-axis, perpendicular to the baseline. The **emitter position $\boldsymbol{E}$** lies at **range $\boldsymbol{R}$** (distance from the array centre to the emitter, metres) in **bearing $\boldsymbol{\theta}$**, where θ is measured *from boresight*, positive toward $+x$ (toward A_0):

$$E = (R\sin\theta,\ R\cos\theta).$$

The emitter transmits a narrowband signal at **carrier frequency $\boldsymbol{f}$** (Hz) with **wavelength $\boldsymbol{\lambda}$** $= c/f$, where $\boldsymbol{c}$ is the **speed of light** (3×10^8 m/s; c is used *only* for the speed of light in this document — the audit's constant phase offset, sometimes called c in code output, is written φ_0 here). At 2.412 GHz, $\lambda = 12.4$ cm; at 5.766 GHz, $\lambda = 5.2$ cm; at 915 MHz, $\lambda = 32.8$ cm. The single most important derived quantity is the **spacing in wavelengths $\boldsymbol{d/\lambda}$** (dimensionless): all ambiguity and sensitivity properties of the array depend on d and λ only through this ratio.

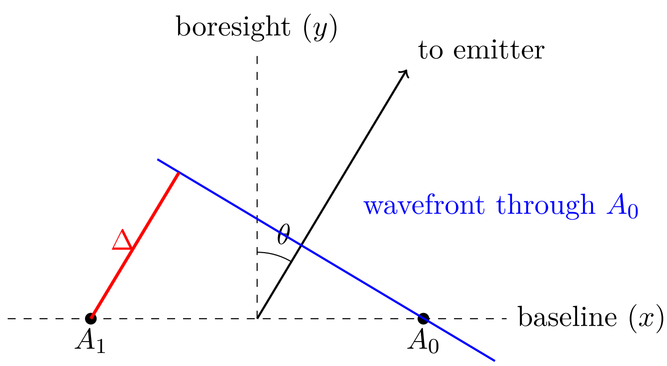

Figure 1: Far-field geometry. The wavefront (perpendicular to the arrival direction) reaches A_0 first; A_1 waits for the wave to cover the extra red path $\Delta = d\sin\theta$.

4.2 Step 1: the far-field assumption

The idea in plain words. Drop a stone in a pond. Near the splash, the ripples are obviously circles. Now stand far away and look at the small patch of water in front of you: the piece of ripple passing you looks like a *straight line*, because you are seeing a tiny arc of an enormous circle. Radio waves spreading out from an emitter do the same thing in 3D: near the emitter the wavefronts

Table 1: Notation used throughout. Symbols are defined in bold where they first appear in the text.

Symbol	Meaning	Units
A_0, A_1	the two receive antennas (channel 0 / channel 1)	—
d	baseline length (antenna spacing)	m
E	emitter position	m
R	range: distance from array centre to emitter	m
θ	emitter bearing, measured from boresight	rad
θ_{mount}	orientation of the array on its platform	rad
f	carrier frequency	Hz
c	speed of light (only this; never a fit parameter)	m/s
$\lambda = c/f$	carrier wavelength	m
d/λ	spacing in wavelengths (drives ambiguity/sensitivity)	—
$\hat{u}$	unit vector from array toward emitter	—
Δ	path-length difference between the two channels	m
$\tau = \Delta/c$	arrival-time difference	s
$x_0[n], x_1[n]$	complex (IQ) baseband samples of the two channels	—
φ	true inter-channel phase difference	rad
$\hat{\varphi}$	estimated phase difference, wrapped to $(-\pi, \pi]$	rad
pi_norm$(\cdot)$	wrap an angle into $(-\pi, \pi]$	—
φ_0	systematic phase offset (audit fit; `offset_c` in code)	rad
g	effective/configured spacing ratio (audit fit)	—
$\Delta\theta$	angular offset: mount rotation / compass bias (audit fit)	rad
$\bar{R}$	circular resultant length (*not* the range R)	—
σ_{circ}	circular standard deviation $\sqrt{-2\ln\bar{R}}$	rad
$\sigma_\varphi, \sigma_\theta$	phase-noise and bearing-noise scales	rad

(surfaces of equal phase — think "crests") are spheres, but by the time they reach a receiver that is far away compared to the size of that receiver, the little piece of sphere crossing the two antennas is as good as flat. "Working in the far field" just means: *pretend the crests are flat sheets travelling in a straight line from the emitter.* The question this subsection answers is: how much error does that pretence introduce for *our* geometry — and the answer will be "about a thousandth of what we care about".

The exact picture. Each antenna measures phase according to its own exact distance from the emitter: $\|E - A_0\|$ for antenna A_0 and $\|E - A_1\|$ for A_1. What the array actually senses (Step 3) is the *difference* of these two distances. So the honest question is: how different is the exact distance difference from the flat-wave answer $d\sin\theta$ that we are about to derive in Step 2?

Doing the algebra once, slowly. Take antenna A_0 at $(+d/2, 0)$ and the emitter at $E = (R\sin\theta, R\cos\theta)$. Pythagoras gives the exact distance:

$$\|E - A_0\| = \sqrt{\left(R\sin\theta - \tfrac{d}{2}\right)^2 + \left(R\cos\theta\right)^2} = \sqrt{R^2 - R\,d\sin\theta + \tfrac{d^2}{4}}\,,$$

where the second equality is just multiplying out and using $\sin^2\theta + \cos^2\theta = 1$. Factor R^2 out of the square root:

$$\|E - A_0\| = R\sqrt{1 \underbrace{-\frac{d\sin\theta}{R} + \frac{d^2}{4R^2}}_{\text{small, because } d \ll R}}.$$

Now use the one approximation tool this whole derivation needs, the square-root shortcut for small x:

$$\sqrt{1+x} \;\approx\; 1 + \frac{x}{2} - \frac{x^2}{8} \qquad (\text{try } x = 0.01 : \sqrt{1.01} = 1.00499,\ \text{and } 1 + 0.005 - 0.0000125 = 1.00499).$$

Applying it with $x = -d\sin\theta/R + d^2/(4R^2)$ and keeping terms up to d^2 (everything smaller is truly negligible) gives, for both antennas at once (upper signs for A_0, lower for A_1):

$$\|E - A_{0,1}\| \;=\; R \mp \tfrac{d}{2}\sin\theta \;+\; \frac{d^2\cos^2\theta}{8R} \;+\; O\!\left(\frac{d^3}{R^2}\right).$$

Read the three terms left to right:

- R: the overall distance. Identical for both antennas, so it cancels completely when we subtract — it carries no direction information.
- $\mp\frac{d}{2}\sin\theta$: opposite sign for the two antennas. Subtracting gives exactly $\Delta = d\sin\theta$ — *this is the flat-wave (far-field) answer of Step 2*, recovered from exact algebra.
- $d^2\cos^2\theta/(8R)$: the first correction — the *curvature* of the wavefront. Crucially it has the *same* sign for both antennas, so it *almost* cancels in the difference; what survives in the difference is bounded by $d^2/(8R)$.

Now plug in our numbers. Wall array, worst case: spacing $d = 5\,\text{cm}$, range $R = 0.9\,\text{m}$, wavelength $\lambda = 12.4\,\text{cm}$ (2.4 GHz). The leftover curvature term is at most

$$\frac{d^2}{8R} = \frac{(0.05\,\text{m})^2}{8 \times 0.9\,\text{m}} = \frac{0.0025}{7.2}\,\text{m} \approx 0.35\,\text{mm}.$$

A third of a millimetre of path error. To turn a path error into a phase error, divide by the wavelength (how many crests fit in it) and multiply by 2π (radians per crest):

$$\text{phase error} \;\le\; \frac{2\pi}{\lambda}\cdot\frac{d^2}{8R} = \frac{2\pi \times 0.00035}{0.124} \approx 0.018 \text{ rad} \approx 1^\circ.$$

Measured phase noise on *good* wall-array datasets is $\sigma_\varphi \approx 0.5\,\text{rad}$ ($\approx 29^\circ$) per window (Section 4.9) — roughly $25\times$ larger. The audit also checked this empirically: near-field curvature was explicitly ruled out as a contributor to the residuals. And the wall array is the *hardest* case; rovers operate at $R \sim 10$–$20\,\text{m}$, where the correction shrinks by another order of magnitude.

Conclusion. From here on we work in the far field: incoming wavefronts are treated as planes perpendicular to the arrival direction, and the error committed is $\sim 1^\circ$ of phase against a $\sim 30^\circ$ noise floor.

4.3 Step 2: path-length difference

Because the wavefronts are planes, the extra distance the wave must travel to reach A_1 after passing A_0 is the projection of the baseline vector onto the arrival direction (Figure 1). Write the unit vector pointing *from the array toward the emitter* as $\hat{u} = (\sin\theta, \cos\theta)$. Then

$$\Delta \;=\; (A_0 - A_1)\cdot\hat{u} \;=\; (d, 0)\cdot(\sin\theta, \cos\theta) \;=\; d\sin\theta.$$

Sanity checks, worth internalizing:

- **Broadside** ($\theta = 0$): $\Delta = 0$. Both antennas sit on the same wavefront; no delay.
- **Endfire** ($\theta = \pm 90^\circ$): $\Delta = \pm d$. The wave travels straight down the baseline; the delay is the full antenna separation.
- The dependence is on $\sin\theta$, not θ: equal increments of angle near broadside change Δ much more than near endfire. This single fact drives the sensitivity analysis of Section 4.7.

4.4 Step 3: from path difference to time delay to phase

The wave covers the extra path Δ at speed c, so channel 1 receives a copy of the signal delayed by

$$\tau = \frac{\Delta}{c} = \frac{d\sin\theta}{c}.$$

For a narrowband signal $s(t)\,e^{j2\pi ft}$ whose complex envelope $s(t)$ varies slowly over τ (always true here: $\tau \le d/c \sim 0.2\,\text{ns}$, versus envelope timescales of $\ge$ tens of ns at our bandwidths), delay is equivalent to a phase rotation of the carrier:

$$x_0(t) = s(t)\,e^{j2\pi ft}, \qquad x_1(t) = s(t)\,e^{j2\pi f(t-\tau)} = x_0(t)\,e^{-j2\pi f\tau}.$$

Define the **measured phase difference** as the phase of channel 1 relative to channel 0:

$$\boxed{\varphi \;=\; \arg\big(x_1 x_0^*\big) \;=\; -2\pi f\tau \;=\; -2\pi\frac{d\sin\theta}{\lambda} \;=\; -2\pi\,\tfrac{d}{\lambda}\,\sin\theta.}$$

This is the entire physical content of a two-element array: *one sinusoid-of-angle, scaled by the spacing in wavelengths.* The sign is a convention (it flips if you swap antenna labels, measure θ from the other side, or conjugate the wrong channel); SPF fixes it by the definition above, and the audit's forward model uses exactly this expression with θ replaced by $\theta - \theta_{\text{mount}}$ (the array's own orientation).

Worked micro-examples with $\frac{d}{\lambda} = \frac{1}{2}$ (the classic half-wavelength array, $\varphi = -\pi\sin\theta$):

θ	$\sin\theta$	φ
0°	0	0
$+30^\circ$	0.5	$-\pi/2 \approx -1.571$
$+90^\circ$	1	$-\pi$
-30°	-0.5	$+\pi/2$
-90°	-1	$+\pi$

Note $\theta = +90^\circ$ and -90° give $-\pi$ and $+\pi$ — the same angle modulo 2π. Even the ideal half-wavelength array is only *just barely* unambiguous.

4.5 Step 4: estimating φ from IQ samples

With sampled complex baseband $x_0[n], x_1[n]$ (Section 3), form the per-sample product

$$z[n] \;=\; x_1[n]\, x_0^*[n], \qquad |z[n]| = A_1[n]A_0[n], \quad \arg z[n] = \varphi + \text{noise}.$$

Each sample is a little arrow whose direction is (a noisy copy of) the quantity we want. The maximum-likelihood-style estimate over a window W is the direction of the *coherent sum* — lay all the arrows tip-to-tail and read off the direction of the total:

$$\hat{\varphi} \;=\; \arg \sum_{n \in W} x_1[n]\, x_0^*[n].$$

("Maximum-likelihood-style": for a single coherent source in independent complex Gaussian noise of constant level, this *is* the ML estimator of φ; under real conditions — AGC, interferers, bursts — it is the sensible default rather than exactly optimal.) Three properties matter in practice:

1. **It is a circular mean.** Averaging $z[n]$ in the complex plane before taking arg handles phase wrapping correctly; averaging raw phases does not ($+179°$ and $-179°$ average to $0°$ instead of $180°$).

2. **It is amplitude-weighted.** Strong samples dominate the sum. Usually good (they have better SNR), but under fast-attack AGC the amplitude is a moving target — gain steps within a window distort the weighting, and gain-index changes can flip channel polarity (a π error on the affected samples).

3. **Windowing trades variance for time resolution.** SPF computes $\hat{\varphi}$ (and a full beamformer, below) over many short windows per capture snapshot rather than one long one, retaining the distributional shape — which the networks consume and the audit summarizes with circular statistics.

One number per window — so where did the ambiguity go? $\hat{\varphi}$ is a single scalar, and it is tempting to think a scalar cannot represent an ambiguous measurement. It can: the ambiguity was never *in* the measurement, it is in the *inversion*. One value of $\hat{\varphi}$, together with $\frac{d}{\lambda}$, implies the *entire* alias set of Section 4.6 — the mirror image and every grating candidate θ_k. Nothing is lost per window under the ideal single-source model: ($\hat{\varphi}$, channel powers) is a sufficient summary of the window, and anyone downstream can expand it into the full multi-peak likelihood at will.

The beamformer: the same information, pre-expanded onto an angle grid. For each candidate angle θ_k on a grid (SPF uses 65 bins), the **beamformer** applies the steering weight $e^{+j2\pi\frac{d}{\lambda}\sin\theta_k}$ and records the coherently combined power. For two elements this reduces, per window, to

$$B(\theta_k) \;=\; P_0 + P_1 + 2\left|\textstyle\sum_W z[n]\right| \cos\!\left(\hat{\varphi} + 2\pi\tfrac{d}{\lambda}\sin\theta_k\right),$$

with P_0, P_1 the window's channel powers — a raised cosine in $\sin\theta_k$, peaked at *every* angle that would explain $\hat{\varphi}$ and dipped at every angle that would not. Note what this formula says: the two-element beamformer is a *deterministic function of* $(\hat{\varphi}, |\sum z|, P_0, P_1)$. Per window, the beamformer and the coherent-sum statistics carry **identical information**.

So can we skip the beamformer? Per window, mathematically, yes. Across windows and across the fleet, practically, no — the beamformer earns its place at *aggregation* time, for four reasons:

1. **Averaging phases across windows is wrong; averaging likelihoods is right.** If half the windows say $\hat{\varphi} = +2.8$ and half say -2.8 (a wrapped, bimodal situation — or two interleaved sources, or burst/junk windows mixed together), the circular mean of the phases is a confident value near $\pm\pi$ that neither population supports being averaged into. Averaging the *beamformer curves* instead keeps both hypotheses alive as two peaks of moderate height. A stack of per-window scalars pushed through a mean destroys multimodality; a stack of per-window likelihood curves preserves it. (SPF feeds the network the windowed stack itself, letting a learned pooling do even better than a plain average.)

2. **A shared angle grid unifies the fleet.** The meaning of $\hat{\varphi} = 1.0\,\mathrm{rad}$ depends entirely on $\frac{d}{\lambda}$, which varies from 0.21 to 1.55 across SPF configurations. On the 65-bin θ grid, "a peak at $\theta = 20°$" means the same thing on every radio, every band, every spacing. The network does not have to learn the arcsin-and-alias expansion separately for every configuration — the beamformer bakes the physics in.

3. **Amplitude and SNR ride along naturally.** Peak height and curve contrast encode $|\sum z|$ and the channel powers — how much this window should be trusted — in the same object, instead of as side channels to be recombined.

4. **Graceful degradation.** An empty or junk window produces a nearly flat beamformer curve ("no opinion"), which averages harmlessly; the same window's $\hat{\varphi}$ is a uniformly random number indistinguishable from a real measurement.

The audit, by contrast, *does* use the scalar path: it has ground truth, works one dataset at a time, and wants residual distributions ($\delta\varphi$ statistics), for which the per-window $\hat{\varphi}$ is exactly the right object. Scalar-per-window and beamformer-per-window are the same physics in two containers; SPF uses whichever container matches the consumer.

4.6 Step 5: inverting phase to angle — and the two ambiguities

Naively inverting the boxed equation:

$$\hat{\theta} \;=\; \arcsin\!\left(-\,\frac{\hat{\varphi}}{2\pi}\,\frac{\lambda}{d}\right).$$

Two distinct ambiguities interfere with this inversion.

Ambiguity 1: front–back (mirror) symmetry. $\Delta = d\sin\theta$ depends only on the projection of the arrival direction onto the baseline. A source at θ and its mirror image at $\pi - \theta$ (same x-component, opposite y) produce *identical* Δ, hence identical φ, for every spacing and wavelength:

$$\sin(\pi - \theta) = \sin\theta.$$

A linear two-element array is physically incapable of telling front from back. SPF therefore works in the folded half-space $\theta \in [-\pi/2, +\pi/2]$ ("reduced theta") at the single-radio level, and resolves the mirror only by fusion: two arrays with different mount orientations on one platform, or one array observed along a trajectory, disagree about the mirror image but agree about the true source.

Ambiguity 2: phase wrapping / grating lobes. The measurable phase is only known modulo 2π: $\hat{\varphi} \in (-\pi, \pi]$ (SPF's pi_norm). The true geometric phase $-2\pi\frac{d}{\lambda}\sin\theta$ ranges over $[-2\pi\frac{d}{\lambda}, +2\pi\frac{d}{\lambda}]$, an interval of total width $4\pi\frac{d}{\lambda}$. Uniqueness of the inversion requires that width to fit inside one 2π turn:

$$4\pi\,\tfrac{d}{\lambda} \le 2\pi \quad \iff \quad \boxed{d \le \frac{\lambda}{2}}$$

— the classical half-wavelength criterion, the spatial twin of Nyquist sampling. When $d > \lambda/2$, all angles

$$\theta_k = \arcsin\!\left(-\frac{\lambda}{d}\left(\frac{\hat{\varphi}}{2\pi} + k\right)\right), \qquad k \in \mathbb{Z},\ \left|\frac{\hat{\varphi}}{2\pi} + k\right| \le \tfrac{d}{\lambda},$$

are exactly consistent with the measurement — the grating lobes. The number of consistent angles is roughly $\lceil 2\frac{d}{\lambda}\rceil$ per half-space.

SPF's fleet spans $\frac{d}{\lambda} \approx 0.21$ to 1.55. At $\frac{d}{\lambda} = 1.27$ (a common 5.8 GHz configuration) a single snapshot admits $\sim$3 candidate bearings per half-space, times the front–back mirror: up to six angles indistinguishable from one measurement. This is *by design*: larger $\frac{d}{\lambda}$ buys angular resolution (next subsection) at the price of ambiguity, and the ambiguity is handed to the learned models and the particle filter, which see many snapshots, two radios, and motion.

4.7 Step 6: sensitivity, resolution, and where the information lives

Differentiate the forward model:

$$\frac{d\varphi}{d\theta} = -\,2\pi\,\tfrac{d}{\lambda}\,\cos\theta.$$

A phase-noise floor σ_φ therefore maps to bearing noise

$$\sigma_\theta \;\approx\; \frac{\sigma_\varphi}{2\pi\,\frac{d}{\lambda}\,|\cos\theta|}.$$

Consequences:

- **Broadside is the sweet spot**; at endfire ($\theta \to \pm 90°$) the derivative vanishes and bearing information disappears entirely — no estimator can fix this.
- **Wider spacing is better, linearly**, wherever the alias candidates can be resolved by other means. Doubling $\frac{d}{\lambda}$ halves σ_θ. This is the quantitative argument for collecting data at $\frac{d}{\lambda}$ well beyond 1/2 and treating disambiguation as a separate, solvable problem.
- With measured wall-array phase noise $\sigma_\varphi \sim 0.5\,\text{rad}$ (corrected residual, good datasets) and $\frac{d}{\lambda} = 0.5$: $\sigma_\theta \approx 0.16\,\text{rad} \approx 9°$ per window at broadside — and windows arrive by the hundreds per second, so the per-snapshot posterior tightens fast if systematics don't bias it.

4.8 Step 7: the real world — a three-parameter systematic model

Real rigs deviate from the boxed model in low-dimensional, physically interpretable ways. The audit fits, per dataset and per channel pair, using ground-truth bearing θ_{gt}:

$$\varphi \;\approx\; \varphi_0 \;+\; g\cdot\Big(-\,2\pi\,\tfrac{d}{\lambda}\Big)\,\sin\!\big(\theta_{\text{gt}} - \Delta\theta\big),$$

with exactly three parameters:

φ_0 **(phase offset; `offset_c` in the audit output):** a constant inter-channel phase — LO distribution, cable-length mismatch, front-end group delay. Harmless to a learned model (it is per-rig constant) but fatal to naive inversion.

g **(gain of the sine, equivalently effective/configured spacing):** $g \neq 1$ means the array behaves as if its spacing were $g\,d$. Causes: mislabeled physical spacing (documented and later hand-fixed for a batch of rover captures: 43 mm labels on a 47 mm array), antenna coupling and connector geometry (the wall fleet shows an effective-spacing floor of ≈6–7 cm at 2.4 GHz regardless of smaller configured values), frequency-dependent electrical length. The audit's receiver-agreement correlation ($\rho \approx 0.64$ between the two independent receivers of the same session) shows g is a real physical property, not fit noise.

$\Delta\theta$ **(angular offset):** mount rotation relative to the platform frame, or — on rovers — compass bias. Era-dependent on rovers (a compass recalibration event visibly moved the fleet-wide bias from -0.15 rad to $+0.04$ rad).

The decomposition matters because the three parameters are *corrections* (deterministic, learnable, even fixable in metadata), while what remains after the fit — the corrected residual spread — is genuine measurement noise. The audit treats them differently: systematic terms feed "label-correction" candidate lists; residual spread feeds quality gates.

4.9 Step 8: quantifying the leftover noise — circular statistics

Residuals $\delta\varphi = \text{pi\_norm}(\hat{\varphi} - \varphi_{\text{model}})$ live on the circle, so ordinary standard deviation is wrong near the wrap. With unit phasors $e^{j\delta\varphi_i}$ and resultant length

$$\bar{R} = \left| \tfrac{1}{N} \sum_{i=1}^{N} e^{j\delta\varphi_i} \right| \in [0, 1],$$

the circular standard deviation is

$$\sigma_{\text{circ}} = \sqrt{-2 \ln \bar{R}}.$$

Interpretation anchors: $\sigma_{\text{circ}} \approx$ the ordinary σ for concentrated data ($\lesssim 0.5$); a uniform (information-free) phase distribution gives $\bar{R} \to 0$ and a finite-N floor of $\sigma_{\text{circ}} \approx 2.1$ at $N = 50$ rising to ≈ 3.0 at $N = 5000$; and mixtures are nonlinear — 30% uniform junk contaminating an otherwise clean channel already pushes σ_{circ} to ≈ 0.9. Heavy tails from AGC gain steps and LNA-bypass polarity flips (π jumps) inflate it further. These anchors calibrate every quality gate in the audit.

4.10 Summary of the chain

$$\theta \xrightarrow{\text{geometry}} \Delta = d \sin\theta \xrightarrow{\tau = \Delta/c} \varphi = -2\pi \tfrac{d}{\lambda} \sin\theta \xrightarrow{\text{rig}} \varphi_0 + g(\cdot) \text{ at } \theta - \Delta\theta \xrightarrow{\text{noise}} \hat{\varphi}$$

Reading the chain left to right generates predictions (the beamformer, the audit's forward model); reading it right to left — through wrapping, aliasing, mirrors, systematics, and noise — is the estimation problem the learned models and filters solve.

5 Approach

The approach has four legs, each matched to a failure mode of the naive inversion of Section 4.6.

5.1 Physics-shaped inputs, learned mapping

Rather than feeding raw IQ or a scalar $\hat{\varphi}$, each snapshot is summarized by the **windowed beamformer**: the 65-bin angular likelihood of Section 4.5, computed over many short windows, plus side information the physics says matters — spacing in wavelengths, carrier frequency, receiver gains, device type, platform type. The network's job is then *not* to rediscover interferometry but to learn the residual map: how a particular rig's beamformer statistics, under AGC and real noise, relate to bearing. Ambiguity is preserved end to end: models output distributions over the 65 bins, not point estimates, and training respects the mirror symmetry of the sensor.

5.2 Two-stage training: single radio, then paired

Stage 1 (single) trains a per-radio network mapping one radio's beamformer statistics to a bearing distribution. **Stage 2 (paired)** loads the frozen stage-1 network (`detach: true`) for each of the two radios on a platform and trains a fusion head that combines them — two differently-oriented arrays disagree on their alias/mirror structure, so the pair resolves much of what a single array cannot. Freezing stage 1 keeps the paired model's improvements attributable to fusion rather than to re-tuned front ends (and a trainer fix now re-freezes correctly across checkpoint resumes).

5.3 Filtering over trajectories

Bearing distributions per snapshot feed particle / Kalman-style filters that integrate over receiver motion, collapsing mirror and grating ambiguities with motion parallax (Section 1). The filter's measurement model is the same forward model as everywhere else; its empirical noise distributions are estimated from held-out data.

5.4 Dataset auditing as part of the method

Because systematics dominate (Section 4.8), the corpus itself is audited with the same physics:

1. For every dataset, predict φ from ground truth via the forward model; fit the three systematic parameters $(\varphi_0, g, \Delta\theta)$; record corrected residual circular spread, NaN fraction, frozen-position tails, timestamp monotonicity, and coverage.
2. Gate into OK / FLAG / QUARANTINE / ERROR with platform-specific thresholds (wall arrays have sub-millimetre truth, rovers carry intrinsic label noise — the same residual means different things).
3. Treat findings by damage type: label-corrupt data is dropped; input-degraded data is A/B tested (it may still teach robustness); feature-mislabeled data is correctable via sidecars.
4. Freeze the historical validation set forever; express quality strata as *named subsets* of it (`val_subset_groups`), so every model past and future is comparable on identical data while new metrics accumulate additively.
5. Run controlled training experiments (identical steps and schedule) on hygiene-filtered training lists to measure — not assume — the value of dropping suspect data.

The audit's key design choice mirrors the modeling philosophy: use the exact physical forward model to define "surprise", fit only low-dimensional, interpretable corrections, and let residual statistics — not intuition — decide what data is trustworthy.

5.5 Status and provenance

All artifacts referenced here are reproducible: the audit PDF from `pdf_scripts/dataset/` (pinned metrics + one command), split manifests from `make_v2_splits.py` (sha256 manifest committed), and experiment configs are append-only copies with provenance-carrying names.